Faculty of Computer Applications & Information Technology

iMCA PROGRAMME

Semester III

Project Report for

242301306 Introduction to Embedded Systems

Project Definition

Electric Self-Defense Shoe

<u>Submitted by:</u>

Group No:16
Aryan Bochiya 202402519010010
Heer Patel 202402519010073
Kathan Patel 202402519010075

ABSTRACT

In today's world, personal safety—especially for women—is a critical concern. The proposed project introduces a wearable self-defense system embedded into footwear that serves two major purposes:

1. Physically deters an attacker through a safe electric shock.
2. Sends an emergency SMS alert to a trusted contact using GSM communication.

This shoe functions through a hidden trigger mechanism that activates a shock delivery system and an automated alert system. The integration of safety and communication in a single wearable device makes it a practical and innovative solution. Its low cost and portability allow it to be used by schoolgirls, working women, and elderly women alike.

OBJECTIVE

The core objective is to design and develop an **Arduino-based self-defense wearable device** that:

- Empowers women with **instant response capability** during emergencies.
- Integrates **electronics safely into everyday footwear**.
- Enables **wireless communication** through GSM modules for remote alerts.
- Ensures the solution is **affordable, durable, and easy to use**.
- Promotes **confidence, independence, and security** for women and girls.

COMPONENTS

Hardware Components:

- ESP8266 MCU
- LCD Display
- High Voltage Generator (Boost Module)
- Relay Module
- Panic Button
- Metal Plate
- Rechargeable Battery
- Wires, PCB, Shoe

Software Components:

- ESP8266 Firmware (Arduino IDE, C/C++)

COMPONENTS FUNCTIONALITY

Hardware Components Functionality:

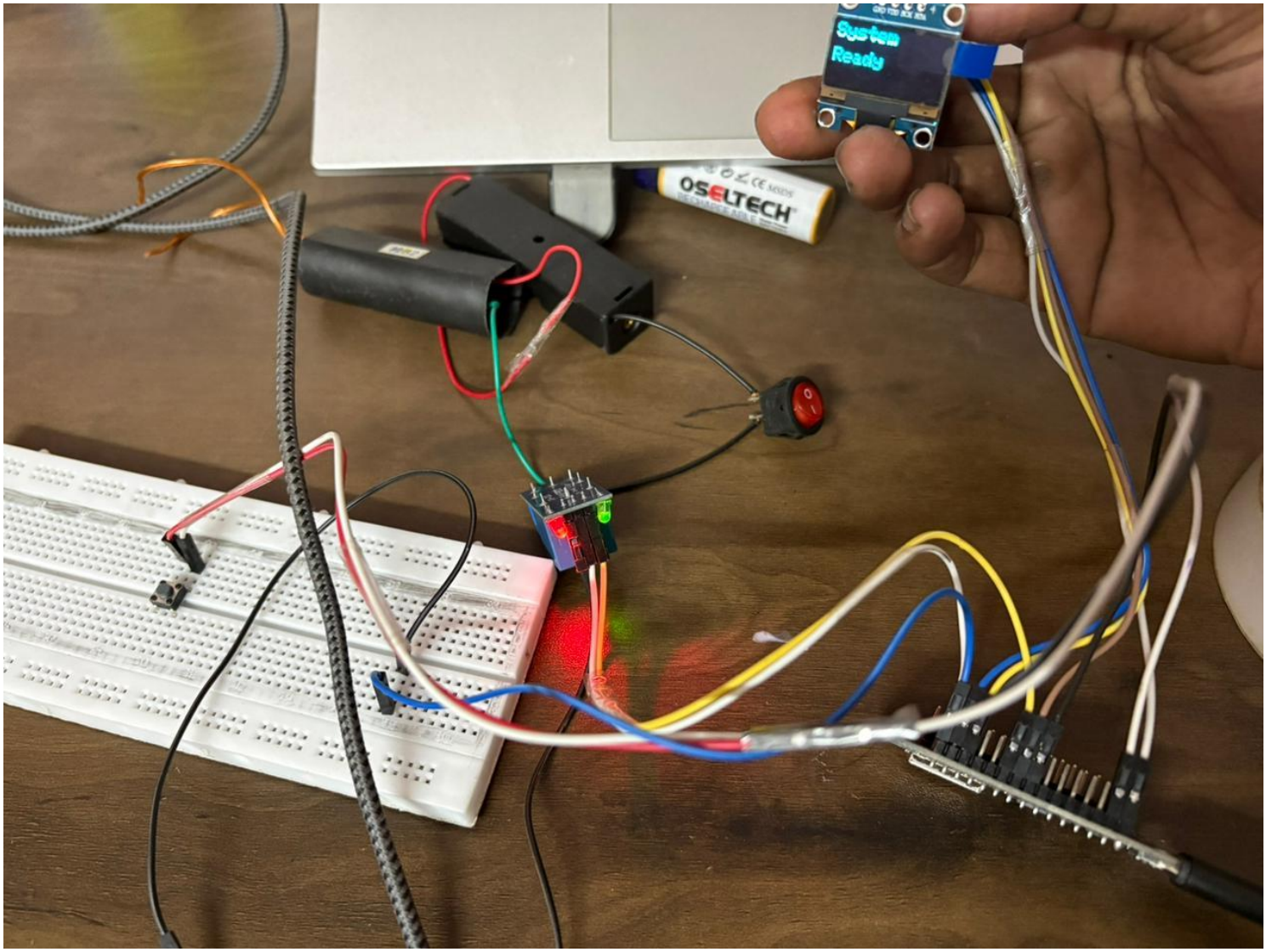
- **ESP8266 MCU – Brain of the system; handles logic, LCD output, and relay control.**
- **LCD Display**- Shows system states: idle, activated, alert sent, or errors (e.g., "No SIM"). Helps the user confirm operation.
- **High Voltage Generator (Boost Module)** – Steps up voltage to produce a strong electric shock for self-defense.
- **Relay Module** – Works like an electronically controlled switch; Arduino turns it ON/OFF to safely control the high-voltage generator.
- **Panic Button** – A hidden emergency button; pressing it signals Arduino to activate the shock system and send alerts.
- **Metal Plate** – Conductive plate that delivers the electric shock when touched by an attacker.
- **Rechargeable Battery** – Portable power source for the entire circuit; ensures the system works even without external power.
- **Wires, PCB, Shoe** – Provide electrical connections, mount the circuit, and conceal the system inside footwear.

Software Components Functionality:

ESP8266 Firmware (Arduino IDE, C/C++):

- Reads panic button input.
- Activates relay to power the HV generator.
- Updates LCD with real-time system status.
- Implements timers, safety checks, and error handling.

CONNECTION DIAGRAM



PROJECT CODE SNIPET WITH COMMENTS TO EXPLAIN CODE SECTION

```
// Debug + Working sketch for NodeMCU + SSD1306 OLED + relay + button
// - OLED address 0x3C (you confirmed it)
// - SDA = D2, SCL = D1
// - Relay: D0 (active LOW assumed: LOW = ON, HIGH = OFF)
// - Button: D5 (wired to GND when pressed, using INPUT_PULLUP)
// Open Serial Monitor at 115200 to see debug logs.

#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_W 128
#define SCREEN_H 64
#define OLED_ADDR 0x3C

Adafruit_SSD1306 display(SCREEN_W, SCREEN_H, &Wire, -1);

#define RELAY_PIN D0
#define BUTTON_PIN D5

const unsigned long ALERT_MS = 3000UL;
const unsigned long DEBOUNCE_MS = 50UL;

bool alertActive = false;
unsigned long alertStart = 0;

// debounce
int lastBtn = HIGH;
unsigned long lastDebounce = 0;
bool handled = false;

void log(const char *s) {
  Serial.println(s);
}

void showReady() {
  display.clearDisplay();
  display.setTextSize(2);
  display.setTextColor(WHITE);
  display.setCursor(0, 0);
  display.println("System");
  display.setCursor(0, 26);
  display.println("Ready");
  display.display();
}

void showAlertScreen() {
```

```
  display.clearDisplay();
  display.setTextSize(2);
  display.setTextColor(WHITE);
  display.setCursor(0,0);
  display.println("!!! ALERT !!!");
  display.setTextSize(1);
  display.setCursor(0, 34);
  display.println("Relay OFF - DANGER");
  display.display();
}

void setup() {
  Serial.begin(115200);
  delay(50);
  Serial.println();
  Serial.println("=== START ===");
  Serial.println("Wire begin SDA=D2, SCL=D1");
  Wire.begin(D2, D1);
  Serial.println("Initializing OLED...");
  if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR)) {
    Serial.println("ERROR: SSD1306 init failed at 0x3C");
    Serial.println("Check wiring: GND->GND, VCC->3V, SCL->D1, SDA->D2");
    for (;;); // stop, we need working display
  }
  Serial.println("OLED OK (0x3C)");
  display.clearDisplay();

  pinMode(RELAY_PIN, OUTPUT);
  pinMode(BUTTON_PIN, INPUT_PULLUP);

  // Start with relay ON (active LOW boards)
  digitalWrite(RELAY_PIN, LOW);
  Serial.println("Relay set to ON (LOW)");
  showReady();
}

void loop() {
  int reading = digitalRead(BUTTON_PIN);

  // debounce logic
  if (reading != lastBtn) {
    lastDebounce = millis();
  }
  if ((millis() - lastDebounce) > DEBOUNCE_MS) {
    if (reading == LOW && !handled && !alertActive) {
      handled = true;
      alertActive = true;
      alertStart = millis();

      Serial.println("Button PRESSED -> entering ALERT state");
      // Turn relay OFF: active LOW boards -> HIGH = OFF
      digitalWrite(RELAY_PIN, HIGH);
      Serial.println("Relay OFF (HIGH)");
```

```
      // Immediately show alert
      showAlertScreen();
    } else if (reading == HIGH) {
      // released -> allow next press
      if (handled) Serial.println("Button released -> ready for next press");
      handled = false;
    }
  }
  lastBtn = reading;

  // if alert active, check timeout
  if (alertActive) {
    if (millis() - alertStart >= ALERT_MS) {
      // restore relay ON
      digitalWrite(RELAY_PIN, LOW); // ON
      Serial.println("Alert period ended -> Relay ON (LOW)");
      alertActive = false;
      showReady();
    }
  }

  // small delay to keep loop responsive and avoid log spam
  delay(10);
}
```

WORKING OF SYSTEM

The project functions based on a combination of **hardware and software triggers** controlled by Esp8266.
Steps:

1. A hidden **panic button** inside the shoe is pressed by the user discreetly.
2. The **ESP8266 detects** this input signal and:
    - Activates a **relay-controlled high-voltage circuit**.
    - Sends power to the **shock module**, which delivers a brief electric pulse through metal electrodes placed in the toe or sole area.

RESULT & ANALYSIS

The project successfully demonstrated that a **wearable embedded system** can integrate *self-defense (shock)* and *communication (SMS alerts)* into a single device. The ESP8266-based system achieved low-cost, portability, and effectiveness, making it a **practical solution for personal safety**. With minor improvements in circuit design and power optimization, the device can be further enhanced for real-world deployment.

THANK YOU!!